International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 8

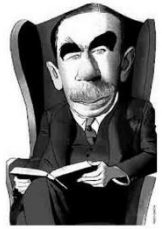
Roadmap

1. Money and prices in the long run
   - Monetary neutrality
   - Evidence
2. Money and FX in the long run
   - Monetary neutrality
   - Exercise
3. Nominal FX in the long run
   - LOP and PPP
   - The Fisher effect
   - A PPP model of long-run FX
   - PPP and Reality

Money and Prices

(in the long run)

Short Run vs. Long Run



- In the short run
  - Assumptions: (i) **prices are sticky**, (ii) output is fixed (for simplicity)
  - The money market equilibrium determines the nominal interest rate

- In the long run
  - Assumptions: **prices are flexible**; output vary but it is determined by real factors (as opposed to monetary policy changes)
  - The money market equilibrium determines the equilibrium price level

Money and Prices in the Long Run

- Money market equilibrium

$$\frac{M^s}{P} = L(\bar{r} + \bar{\pi}, \bar{Y})$$

  - $\bar{Y}$ is the level of output determined by real factors (labor force, capital stock, technology, etc.)
  - $\bar{r}$ is the real interest rate determined by real factors (trend output growth, population growth, etc.)
  - $\bar{\pi}$ is the long-run inflation (inflation target by the central bank)
  - **Bars on top of the variables** mean that the variables are **exogenous**. In this case, it means that they do not depend on the level of money supply

Money and Prices in the Long Run

- Long-run monetary neutrality (proposition)
  - **Real output**: a change in the level of the supply of money does not affect the long-run value of real output
  - **Price level**: a *permanent* increase in the money supply causes a proportional increase in the price level's long-run value

- Changes in money demand also affect the long-run price level
  - $P = M^s/L(\bar{R}, \bar{Y})$ which implies $\Delta P/P \approx \Delta M^s/M^s - \Delta L/L$

Empirical Evidence

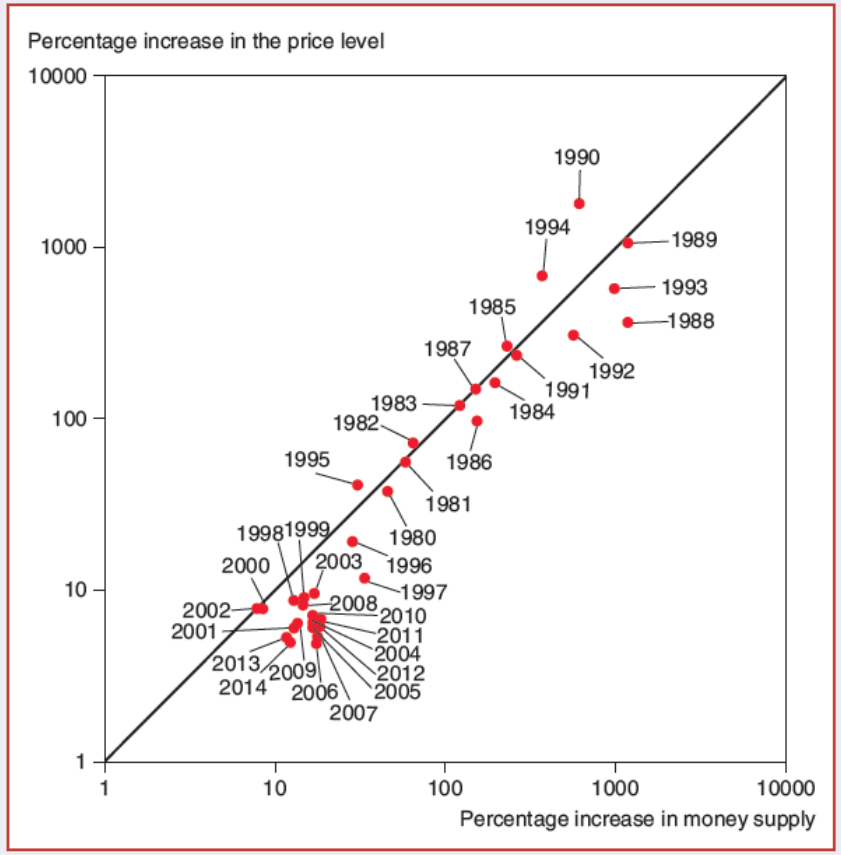


FIGURE 15-10

Average Money Growth and Inflation in Western Hemisphere Developing Countries, by Year, 1980–2014

Even year by year, there is a strong positive relation between average Latin American money supply growth and inflation. (Both axes have logarithmic scales.)

Source: World Bank development indicators database and own calculations. Regional aggregates are weighted by shares of dollar GDP in total regional dollar GDP.

1. Money supply and price level
   Over a long period (1980-2014), changes in the price level are highly correlated with changes in the money supply.

2. Money demand and price level
   The deviations from the 45-degree line suggest that changes in money demand also affect the price level.

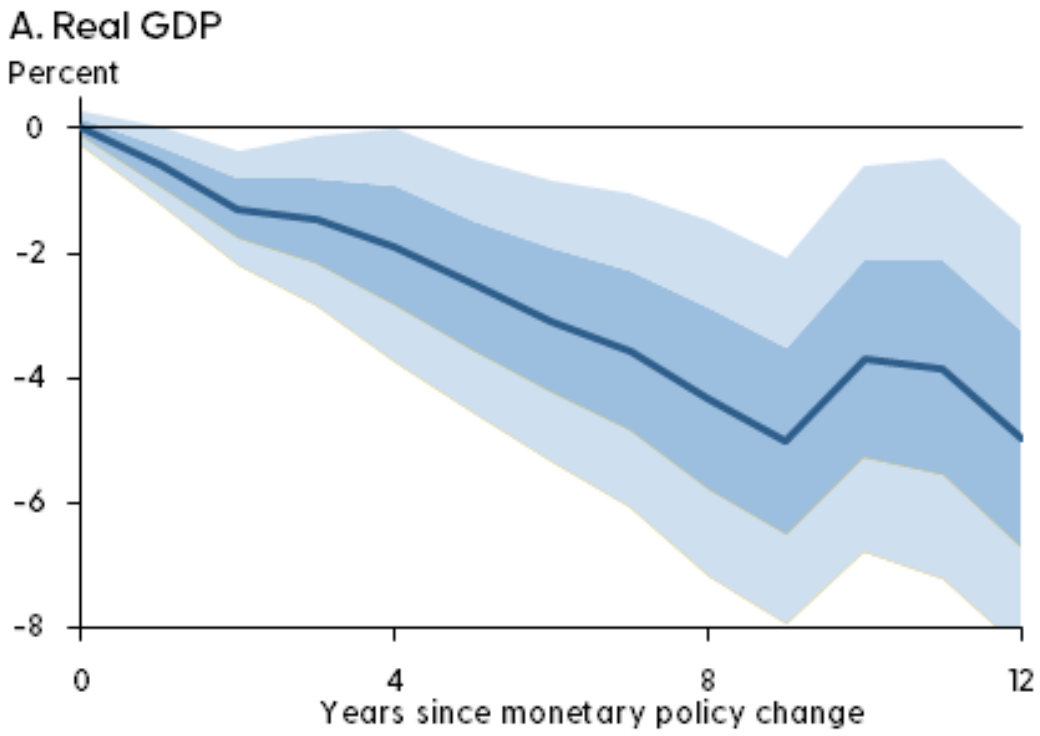
Is Money Really Neutral in the Long Run?

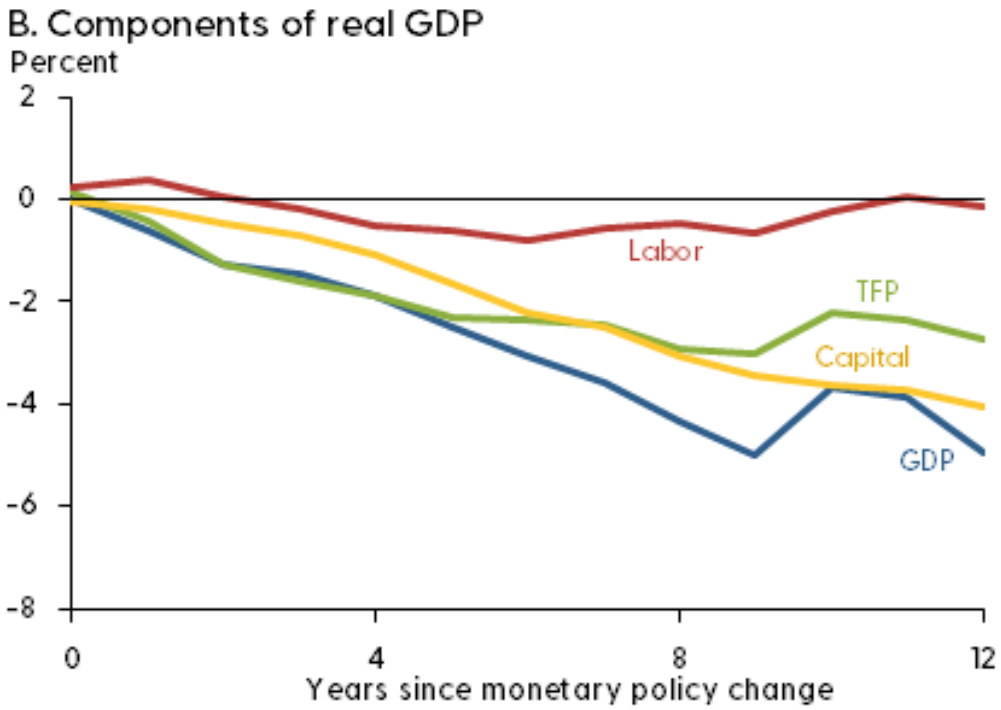
Effects of Monetary Policy in the Long Run

- Why can one expect money ***not*** to be neutral in the long run?
  - R&D reaction to interest rate changes
  - Human capital depreciation during recessions

- Jordà, Singh, Taylor (2023) test the long-run neutrality of MP
  - Historical data: 17 advanced economies over the period 1900 to 2015, excluding the two World Wars
  - Need a good identification of monetary policy shocks
    - Identification strategy: countries that peg their currencies to the US dollar must react to the US monetary policy variations unrelated to economic activity in those countries

Jordà, Singh, Taylor (2023)





Note: Responses to a 1% unexpected change in domestic short-term interest rate. Full sample: 1900–2015, World Wars excluded.
In panel A, darker shading indicates one-standard-error and lighter shading indicates two-standard-error confidence bands around average.
See Jordà, Singh, and Taylor (2023) for estimation details

Evidence on the Innovation Channel

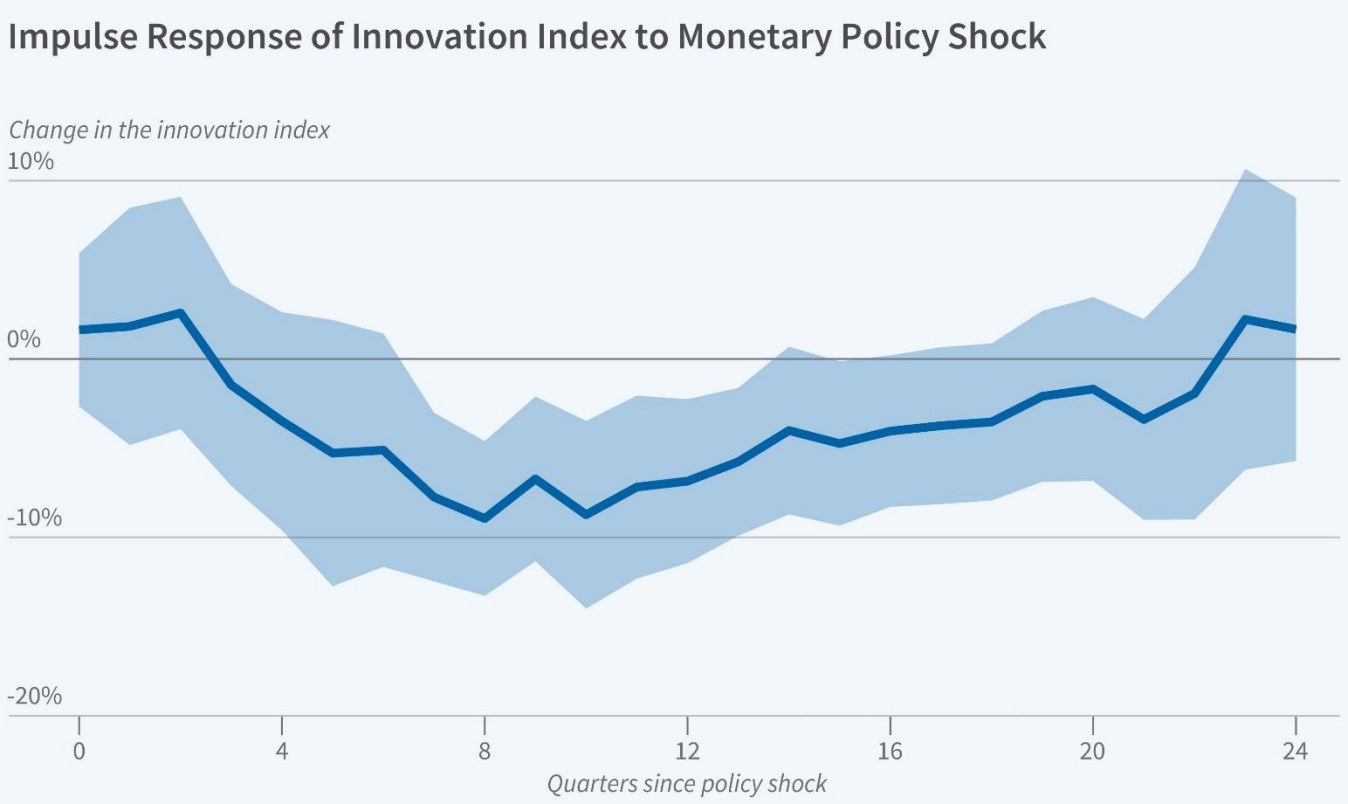


Figure shows response to a 100 basis point interest rate increase from a monetary policy shock. Shaded area represents 90% confidence intervals.
Source: Researchers' calculations using data from NBER Working Paper w17769.

Source: Ma, Zimmermann (2023)

Money and FX

(in the long run)

Money and FX in the Long Run

- Long-run monetary neutrality (proposition) **again**
  - a permanent increase in a country's money supply causes a *proportional* long-run depreciation of its currency against foreign currencies

$$E = \frac{P}{P^*}$$

- Why?
  - Purchasing power parity theory (next time)

Short Run vs. Long Run

- In the short run, when prices (and output) are fixed
  - changes in money supply affect interest rates and, as a result, the exchange rate

$$M \uparrow \Rightarrow \boldsymbol{R} \downarrow \Rightarrow E \uparrow$$

- In the long run, when prices are flexible
  - changes in money supply affect prices and, as a result, the nominal exchange rate

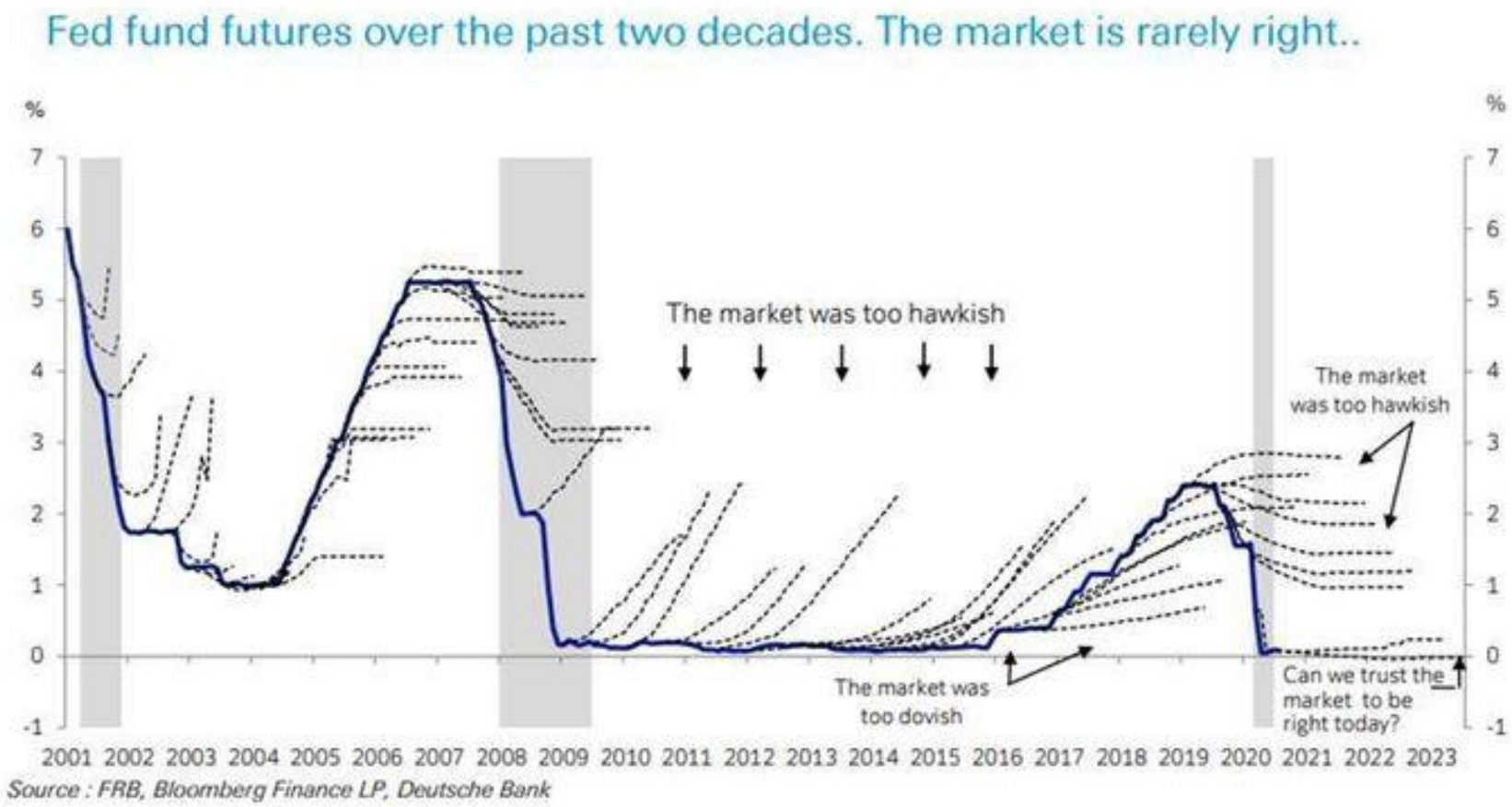
$$M \uparrow \Rightarrow \boldsymbol{P} \uparrow \Rightarrow E \uparrow$$

Exercise

A **permanent** increase in domestic money supply

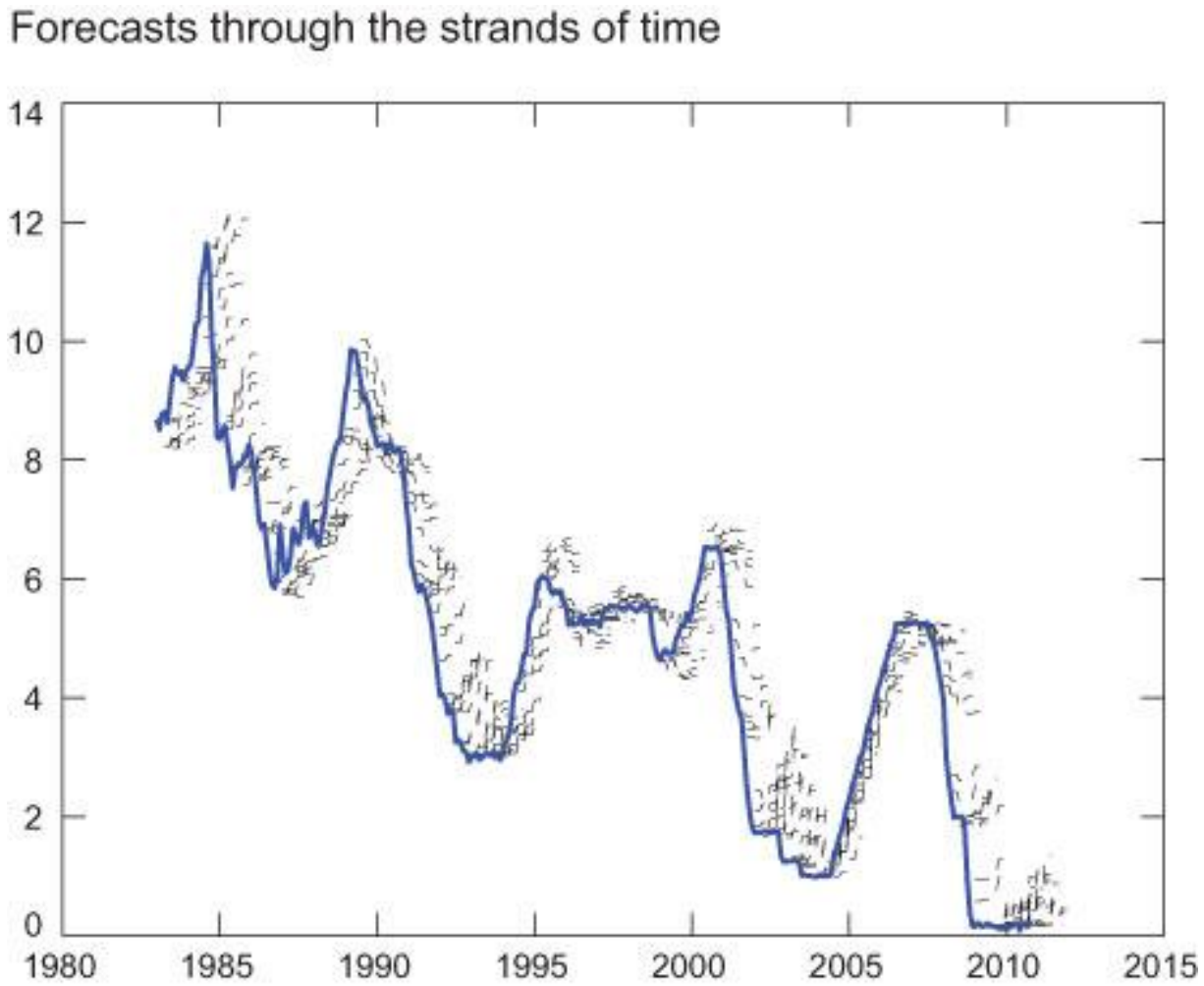
- How does the current exchange rate change when a domestic central bank increases its money supply **permanently**?

Markets Expectations about Monetary Policy



Source: [MarketWatch]

Markets Expectations about Monetary Policy



Sources: Blue Chip Financial Forecasts Survey; authors' calculations.

Source: [Liberty Street Economics]

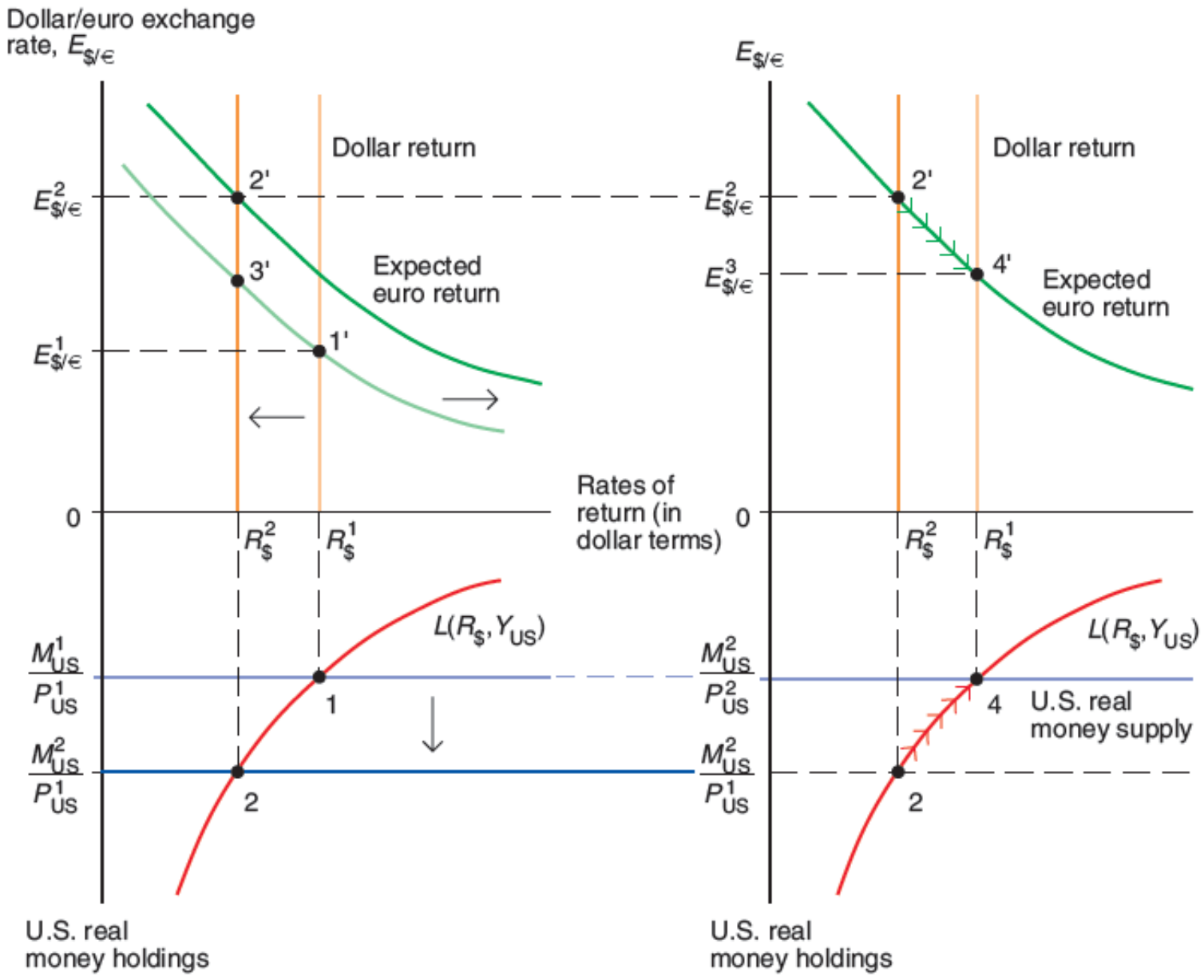
Exercise

A **permanent** increase in domestic money supply

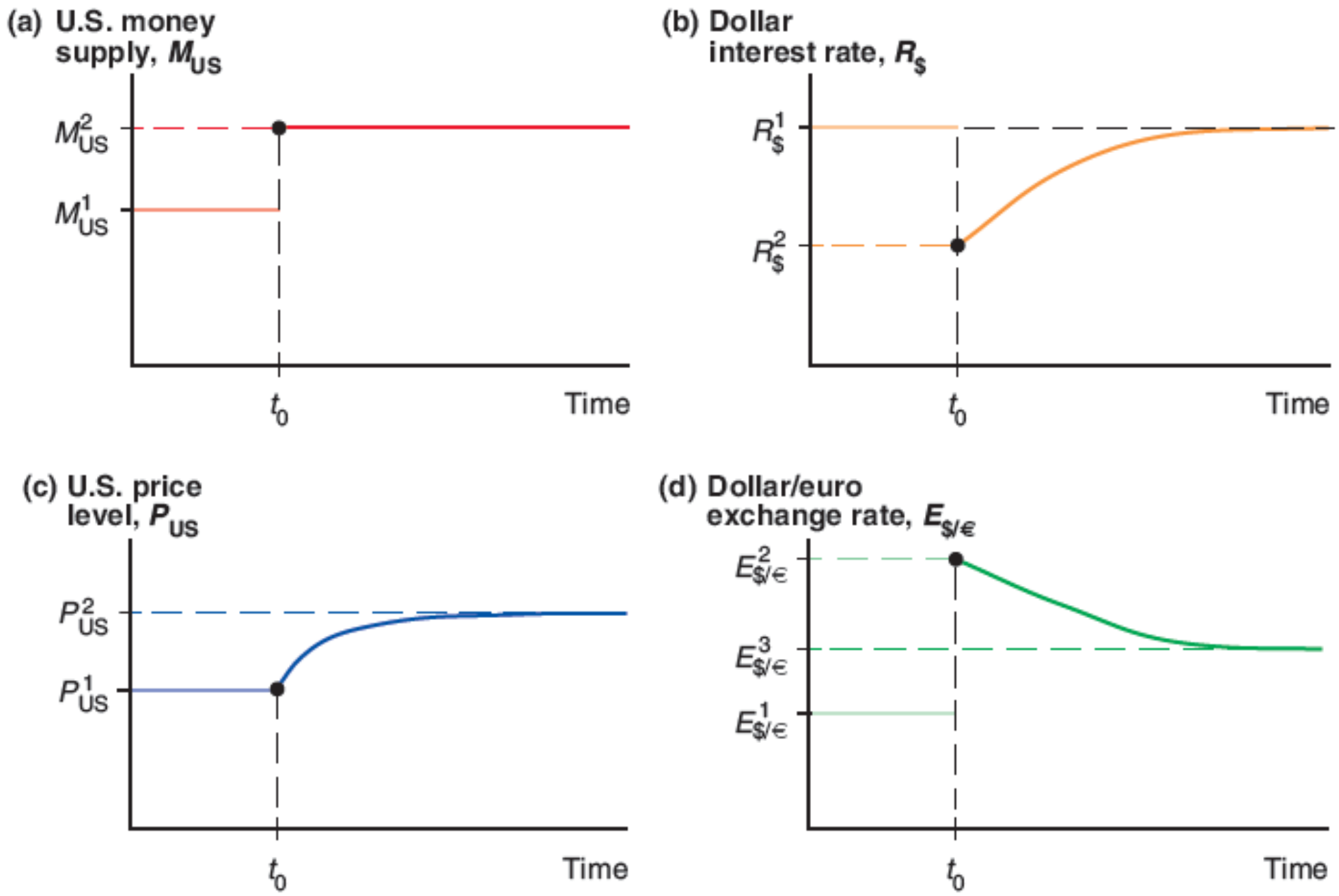
- This exercise combines our short-run and long-run analysis of monetary changes
  - A change in money supply **not only** affects the nominal interest rate and exchange rate like in our previous short-run analysis, **but also** the expected exchange rate

- It is sometimes called

  "the Dornbusch overshooting model"



(a) Short-run effects **(b) Adjustment to long-run equilibrium**


(a) U.S. money supply, M_{US}
M^2_{US}
M^1_{US}
t_0
Time
(b) Dollar interest rate, $R_\$$
$R^1_\$$
$R^2_\$$
t_0
Time
(c) U.S. price level, P_{US}
P^2_{US}
P^1_{US}
t_0
Time
(d) Dollar/euro exchange rate, $E_{\$/€}$
$E^2_{\$/€}$
$E^3_{\$/€}$
$E^1_{\$/€}$
t_0
Time

Exercise Takeaways

A **permanent** increase in domestic money supply

1. **We say that** the exchange rate **overshoots** when its immediate response to a disturbance is greater than its long-run response

2. The model helps explain why exchange rates move so sharply in the short run

3. **Overshooting** is a direct result of a sluggish short-run price level adjustment and the interest parity

Nominal FX in the long run

LOP and PPP

The Law of One Price (LOP)

- In the long run, it is reasonable to assume that identical goods sell for the same price (after conversion to the same currency) in each country

$$E_{\$|€} \cdot P_E^i = P_{US}^i$$

  – *i* is a good index
  – $E_{\$|€}$–nominal exchange rate
  – P_{US}^i– US price of good *i*
  – P_E^i– European price of good *i*

Purchasing Power Parity (PPP)

- The PPP theory proposes

$$E_{\$|€} = P_{US}/P_E$$

  - $E_{\$|€}$–nominal exchange rate
  - P_{US}– domestic price level
  - P_E – foreign price level

David Ricardo
(1772-1823)

LOP vs PPP

- $E_{\$|€} \cdot P_E^i = P_{US}^i$ for all *i* $\Rightarrow$ $E_{\$|€} \cdot P_E = P_{US}$
  - The statement requires domestic and foreign consumption baskets to be identical (Note true in reality but we will abstract from this for now).

- $E_{\$|€} \cdot P_E = P_{US}$ $\not\Rightarrow$ $E_{\$|€} \cdot P_E^i = P_{US}^i$ for all *i*
  - For PPP to hold, the LOP is not necessary

- Proponents of PPP hold that it is useful for understanding of long-run FX movements even though the LOP may not hold

Absolute and Relative PPP

- Absolute PPP is what we so far called PPP

$$E_{\$|€} = P_{US}/P_E$$

- Relative PPP (weaker than absolute PPP)

$$\frac{E_{\$|€,t} - E_{\$|€,t-1}}{E_{\$|€,t-1}} = \pi_{US,t} - \pi_{E,t}$$

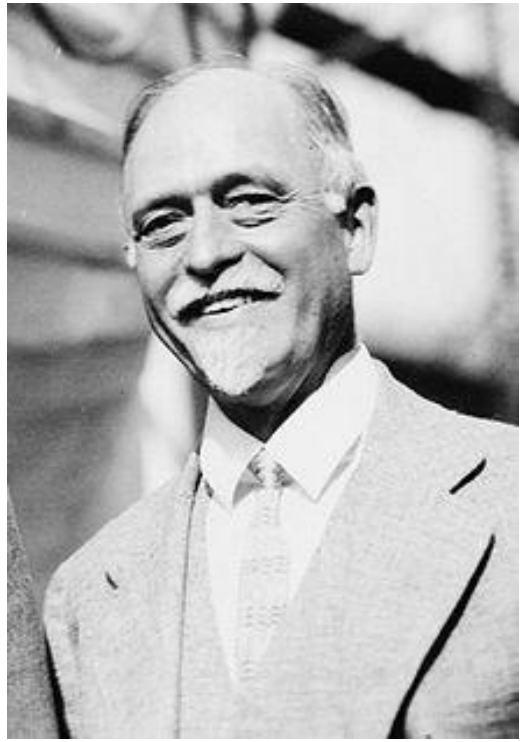
  – $\pi_{US,t} = (P_{US,t} - P_{US,t-1})/P_{US,t-1}$ is inflation in the US, $\pi_{E,t}$ - in Eurozone

- Issue
  – Absolute PPP needs identical baskets of goods in both countries to hold
  – Relative PPP likely overcomes the different baskets issue

The Fisher Effect

Irving Fisher (1867-1947)

- Contribution to economics
  - The interest parity, the Fisher effect, the theory of debt deflation, etc.
  - "the greatest economist the United States has ever produced“ (Milton Friedman)
- Predictions
  - Oct 15, 1929: “Stock prices have reached what looks like a permanently high plateau”
  - Oct 28-29, 1929: Dow Jones Industrial average falls by 25%



The Fisher Effect

- **UIP**

$$R - R^* = \frac{E^e - E}{E}$$

- Assume that financial markets expect **(relative) PPP** to hold, that is,

$$\frac{E^e - E}{E} = \pi^e - \pi^{*e}$$

- Combine

$$R - R^* = \pi^e - \pi^{*e}$$

- The Fisher effect: *a rise in the domestic inflation rate causes an equal rise in the interest rate on deposits of domestic currency* ***in the long run****, when other factors remain constant*

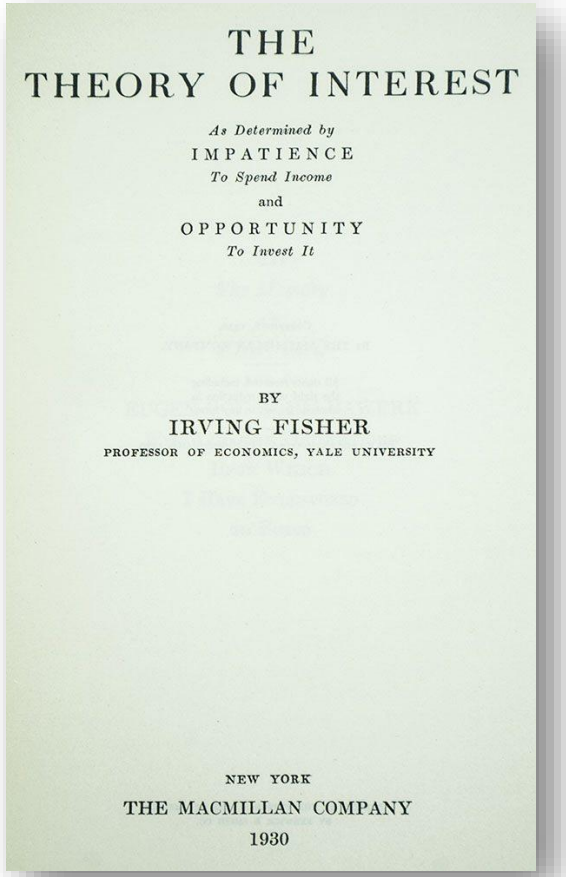
The Fisher Effect

- UIP + relative PPP

$$R - R^* = \pi^e - \pi^{*e}$$

- Note that

$$R - \pi^e = R^* - \pi^{*e} \Rightarrow \boldsymbol{r = r^*}$$


THE
THEORY OF INTEREST

As Determined by
IMPATIENCE
To Spend Income
and
OPPORTUNITY
To Invest It

BY
IRVING FISHER
PROFESSOR OF ECONOMICS, YALE UNIVERSITY

NEW YORK
THE MACMILLAN COMPANY
1930

A PPP Model of Long-Run FX

(A PPP Model of FX under Flex Prices)

A PPP Model of Long-Run FX

NOTA BENE

$$E_{\$|€} = P_{US}/P_E$$

$$P_{US} = \frac{M_{US}^s}{L(\bar{r}_{US} + \bar{\pi}_{US}, \bar{Y}_{US})}$$

$$P_E = \frac{M_E^s}{L(\bar{r}_E + \bar{\pi}_E, \bar{Y}_E)}$$

Predictions

1. $M_{US}^s \uparrow \Rightarrow P_{US} \uparrow, E_{\$|€} \uparrow$
2. $\bar{Y}_{US} \uparrow \Rightarrow P_{US} \downarrow, E_{\$|€} \downarrow$
3. $(\bar{r}_{US} + \bar{\pi}_{US}) \uparrow \Rightarrow P_{US} \uparrow, E_{\$|€} \uparrow$ (in contrast to our short-run analysis)

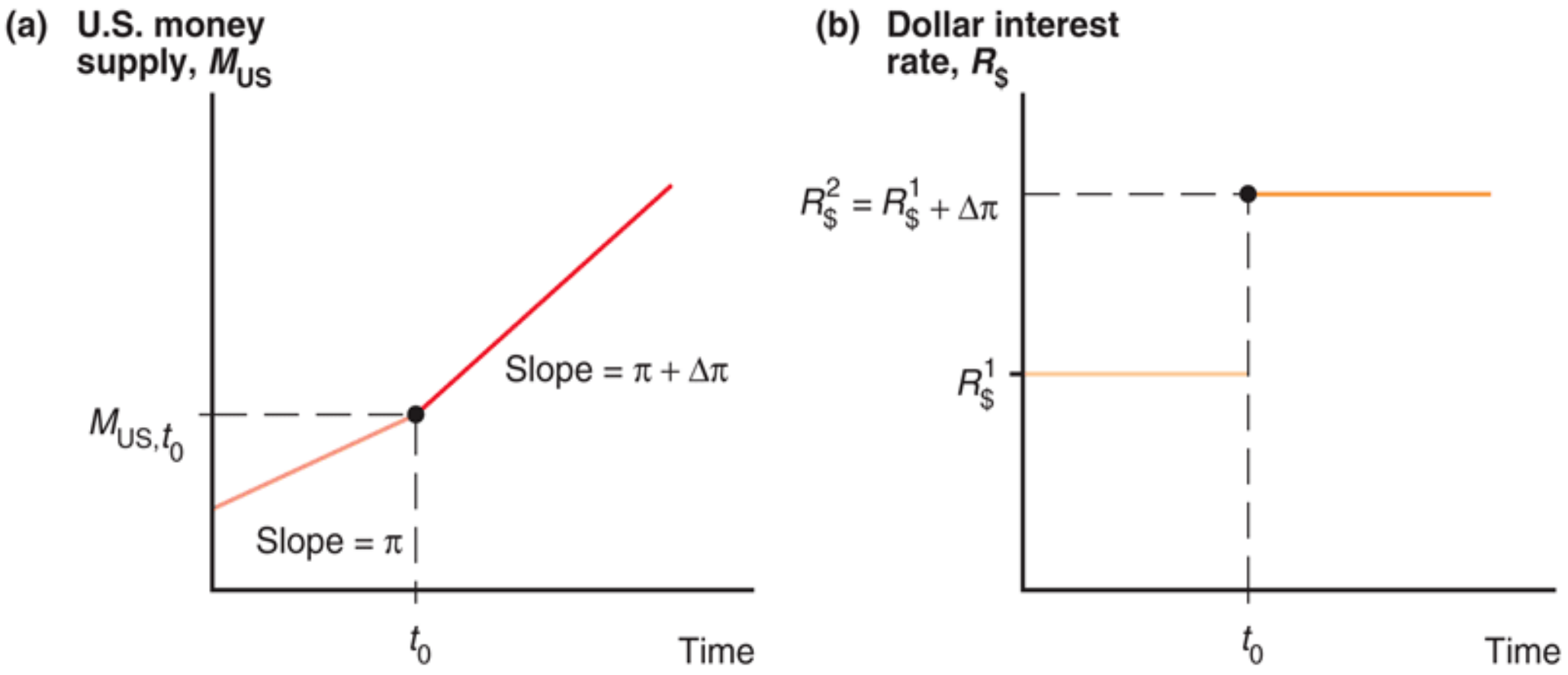
Exercise

A permanent increase in money **growth rate**

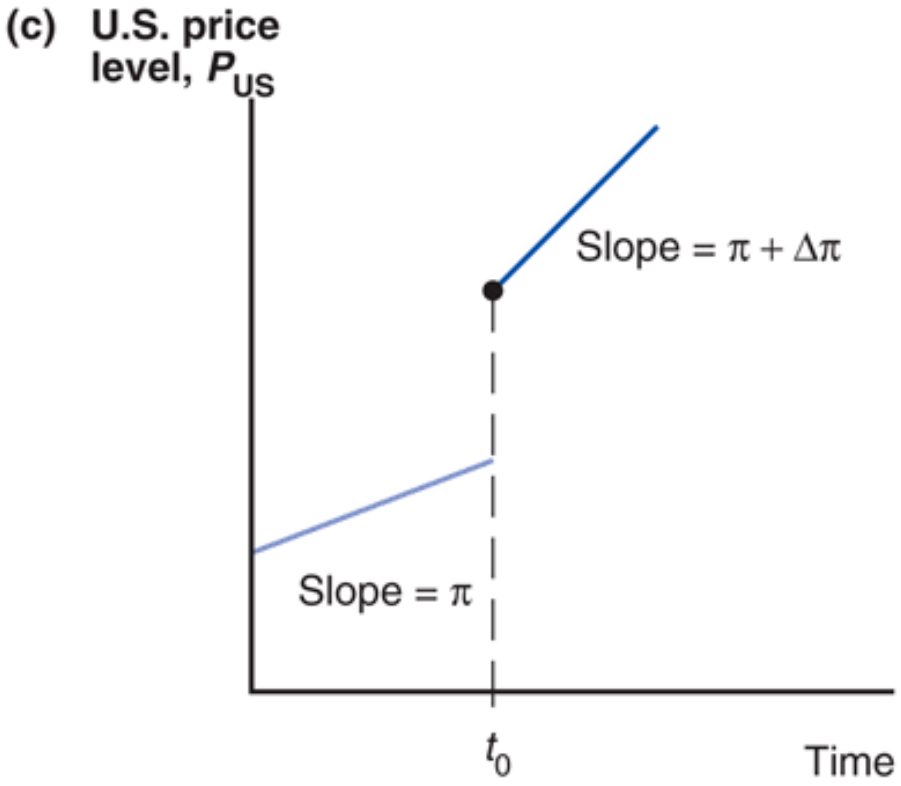
- Home central bank increases the growth rate of the money at time t_0 from π to $\pi + \Delta\pi$ after t_0 (the foreign inflation rate remains the same)
  - For inflation to adjust "fast," we assume that we consider long periods. For example, t_0 is 2019*, while* t_0+1 is 2022.

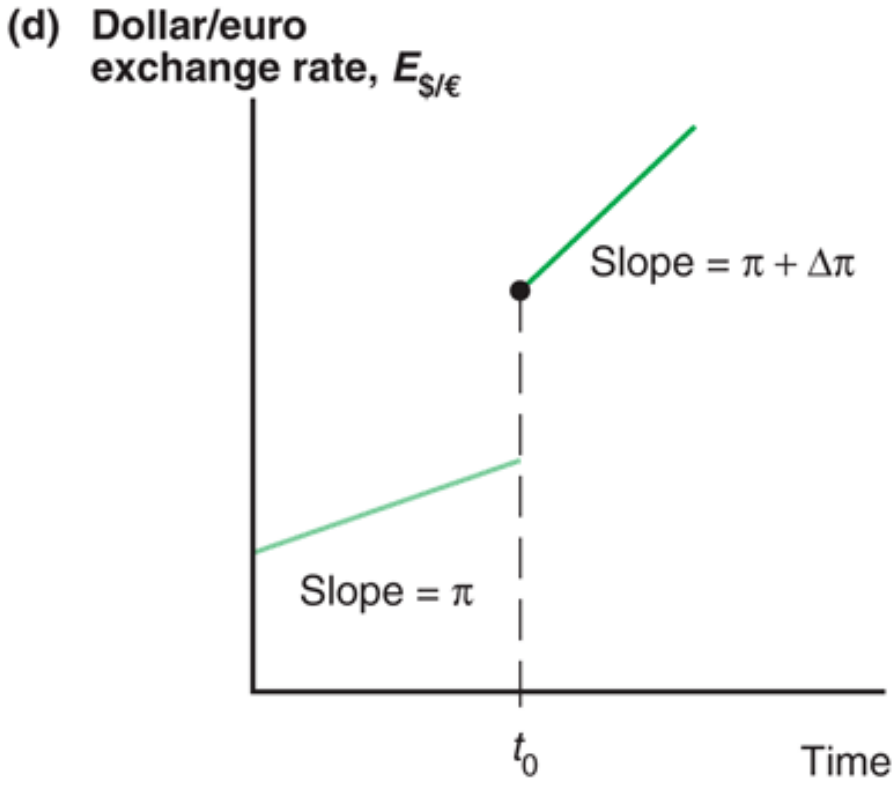
- **According to the Fisher effect**, the domestic interest rate will increase to a new higher level

A Permanent Increase in Money Growth



A Permanent Increase in Money Growth





$$P = \frac{M^S}{L(R, Y)}$$

$$E = \frac{P}{P^*}$$

A Key Takeaway of the Exercise

- To understand how nominal interest rate variations affect the exchange rate, one has to take into account the factors that caused the interest rate change

Readings

- Next lecture:
  - KOM, Chapter 16